

Length Sensitivity of Learned MT Metrics: Encoder Alignment, Domain Adaptation, and Training-Mix Composition

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August 2026 — working report

Abstract

Learned MT metrics such as COMET are trained on single sentences. In practice they are applied to paragraphs and documents. We study what this mismatch costs, in three steps. First, we isolate length itself. We build blocks of k consecutive FLORES+ sentences and ask whether encoders still separate a translation from a copy with a single injected error. COMET’s encoder drops below chance already at $k=3$; sentence aligners (LaBSE, E5) degrade far more slowly. A matched-core design shows the drop persists with meaning-free filler, so it is a property of the representation, not of content. Second, we revisit domain adaptation on Bio-MQM. Fine-tuning WMT22-COMET-DA in domain moves Pearson slightly (+0.018) and rank correlation not at all, and leaves encoder representations essentially unchanged. Third, we continue WMT22-COMET-DA and WMT22-COMETKIWI-DA on WMT training mixes that vary only the fraction of single-sentence data, at constant total size. We document a subtle training-pipeline failure that invalidated a first 36-arm sweep, and a corrected protocol with an identical budget in every arm. We evaluate the retrained metrics on sentence-level, aggregated, and natively long test sets, on MetaDocEval discourse perturbations, and on the block-alignment task. In the first corrected wave ($f \in \{0, 100\}\%$), every arm beats its base metric at every window size, with no catastrophic forgetting; the long-only arm is best on native documents and the sentence-only arm on single sentences, but the composition effect is small. MetaDocEval accuracy does not rise with context for any arm: the gains are score-level, not discourse-level. An audit of the long-text data path found one genuine defect: CometKiwi’s concatenated input silently truncates 7–11% of the long training rows, an asymmetry to keep in mind for QE-vs-DA comparisons.

1 Introduction

COMET-style metrics dominate MT evaluation. They are regressors on top of a multilingual encoder, trained on human judgments of single sentences. Yet users score paragraphs and documents with them. Three questions follow.

Q1. Does input length *by itself* degrade the encoder signal that the metric relies on? We answer this with controlled alignment tasks on FLORES+ (Section 2).

Q2. What does fine-tuning on a new domain actually change? We reproduce the Bio-MQM setting of Zouhar et al. (2024) (Section 3).

Q3. If we continue training the metric on longer inputs, what does that buy, and what does it cost at the sentence level? We run a controlled sweep over the sentence fraction of the training mix (Section 4).

Throughout, “COMET” means WMT22-COMET-DA (Rei et al., 2022a) and “CometKiwi” means the reference-free WMT22-COMETKIWI-DA (Rei et al., 2022b). Both use an XLM-R large encoder (Conneau et al., 2020).

encoder	xsim++ error ↓				detection rate ↑ (chance .50)			
	$k=2$	$k=3$	$k=4$	$k=5$	$k=2$	$k=3$	$k=4$	$k=5$
COMET encoder	.72	.87	.94	.98	.59	.47	.40	.34
+ Bio-MQM ft.	.72	.87	.94	.98	.58	.46	.39	.34
XLM-R mean	.87	.93	.97	.98	.51	.48	.46	.44
LaBSE	.27	.48	.63	.76	.93	.89	.87	.83
E5	.24	.44	.60	.73	.94	.91	.89	.85

Table 1: Block-xSIM++ on FLORES+, mean over de/es/fr/ru. One injected error per block. COMET’s encoder loses the error as k grows; aligners keep it.

2 Q1: length in isolation — block alignment on FLORES+

2.1 Setup

FLORES+ (NLLB Team, 2022; Burchell et al., 2024) provides sentence-aligned articles in many languages. A *block* is k consecutive sentences of one article, concatenated; blocks do not overlap. We use $k \in \{2, 3, 4, 5\}$, languages de/es/fr/ru with English pivot, direction en→xx. Per language there are 846 blocks at $k=2$ and 141 at $k=5$.

We inject exactly one error into one sentence of a block, following the xSIM++ taxonomy (Chen et al., 2023): causality (negation, antonym, modality), entity replacement, and number changes. Each perturbed block differs from its source block by a single edit.

We compare five frozen encoders by cosine similarity: COMET’s encoder, the same encoder after Bio-MQM fine-tuning (Section 3), mean-pooled XLM-R large, LaBSE (Feng et al., 2022), and multilingual E5 (Wang et al., 2022). Three probes separate the failure modes:

- **block-xSIM++:** retrieval over the full pool of true and perturbed blocks. *xsim* error uses true blocks only (pure alignment); *xsim++* error adds the perturbed copies; the *detection rate* is $P[\cos(q, \text{gold}) > \cos(q, \text{perturbed})]$, chance 0.5.
- **fixed-pool duel:** every retrieval is one gold block against exactly D single-error negatives, averaged over all subsets in closed form. Task difficulty is identical at every k ; only length varies. Chance error is $D/(D+1)$.
- **matched core:** the *same* core sentence (with its perturbed variants) is embedded in filler of total length $L \in \{60, 120, 240, 480\}$ XLM-R tokens. Filler types: *inert* (a repeated neutral phrase), *neutral* (unrelated FLORES sentences), *distractor* (same-article sentences), *natural* (contiguous corpus text). Any change across L is attributable to length, never to content. All inputs stay under the 512-token limit, so truncation plays no role.

2.2 Results

Alignment is easy; error detection is not. Classic *xsim* error is near zero for COMET, LaBSE and E5 at every k . The metrics-relevant signal is elsewhere: with perturbed copies in the pool, COMET’s retrieval collapses (Table 1). Its detection rate falls below chance at $k=3$. The dedicated aligners degrade too, but stay far above chance. Bio-MQM fine-tuning changes nothing.

The pool is not the cause. In block-xSIM++ the candidate pool grows with k . The duel removes this confound: with a fixed number of negatives per decision, COMET is at or below chance at every k and still degrades monotonically (duel error at $D=4$: $.79 \rightarrow .95$ from $k=2$ to $k=5$; chance .80). LaBSE and E5 sit far above chance at all k ($.27 \rightarrow .53$ and $.22 \rightarrow .48$).

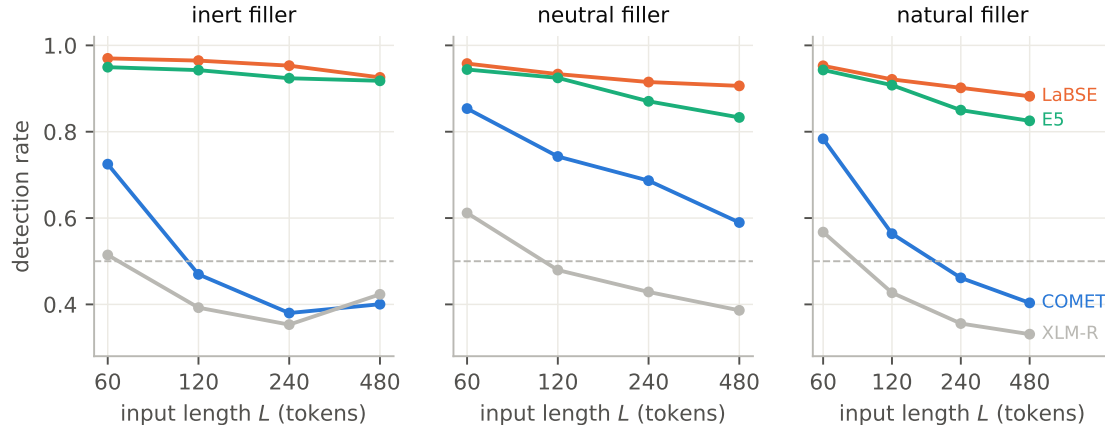


Figure 1: Matched-core detection vs. total input length L (XLM-R tokens), per filler type, mean over de/es/fr/ru. The same core sentence and the same single error at every L ; only filler length varies. Dashed line: chance.

Length alone is sufficient. The matched-core results (Figure 1) make the strongest point. With *inert* filler — a repeated, meaning-free phrase — COMET’s detection falls from .73 at $L=60$ to .47 at $L=120$, below chance. No semantics is added; only length. With natural filler the picture is the same (.78 $\rightarrow$.40). LaBSE never falls below .88 and E5 never below .82, at any L and filler type. As a scalar summary, COMET loses 0.11 detection per doubling of L with inert filler (LSI = $-.105$, 95% CI $[-.115, -.095]$) and 0.12 with natural filler; LaBSE and E5 lose at most 0.02–0.04 and never cross the .75 mark that defines L_{50} . A layer-wise probe of XLM-R shows the error signal is progressively lost toward the top layers and pooling; with natural filler it is still present in the early and middle layers, while with inert filler it fades earlier. The top-layer embedding of the whole input barely moves when the core is perturbed (cosine $\geq .97$ to the clean version): the negatives are geometrically almost indistinguishable.

Reading. Sentence-level error sensitivity of COMET’s encoder does not survive concatenation. The failure is representational and driven by length itself. This motivates Q3: can training on longer inputs restore the signal?

3 Q2: domain adaptation on Bio-MQM

This part is secondary; it reproduces the setting of Zouhar et al. (2024) and provides the fine-tuned checkpoint used above.

3.1 Setup

Bio-MQM provides sentence-level MQM annotations in the biomedical domain, 10 language pairs. We map error severities to penalties (minor -1 , major -5 , critical -25), average over annotators, z-normalise per language pair and squash by a sigmoid. The paper’s dev documents form our training set, its test documents our validation set. We continue WMT22-COMET-DA with a conservative schedule (encoder LR 5×10^{-7} , head LR 10^{-5} , effective batch 64).

3.2 Results

Fine-tuning helps calibration a little and ranking not at all (Table 2): mean Pearson $+0.018$, mean Kendall -0.002 . Representation probes tell the same story. Frozen-probe accuracies on XNLI, PAWS-X and news classification, and full fine-tune accuracies on XNLI and PAWS-X,

LP	Pearson		Kendall τ	
	base	fine-tuned	base	fine-tuned
de-en	.279	.339	.354	.352
en-de	.485	.560	.450	.454
en-es	.293	.330	.329	.326
en-fr	.433	.494	.386	.382
en-ru	.571	.525	.351	.352
es-en	.451	.444	.343	.340
fr-en	.465	.459	.347	.346
ru-en	.470	.437	.288	.279
mean	.431	.448	.356	.354

Table 2: Sentence-level correlation with human MQM on Bio-MQM test documents: WMT22-COMET-DA before and after in-domain fine-tuning.

match the base COMET encoder within ± 0.01 . What moves representations is COMET training itself, not the domain adaptation on top (frozen-probe XNLI: XLM-R .393, COMET .445). This is consistent with the paper’s core finding that fine-tuned metrics gain little robustness from such adaptation.

Open items. zh pairs are not yet in our correlation runs. A side-by-side table against the published per-LP numbers, under the paper’s exact score aggregation, remains to be built. WMT25 provides additional biomedical-adjacent data that could extend this comparison.

4 Q3: training-mix composition at constant budget

4.1 Data: three pools, one schema

We build three training pools with a common COMET schema (src, mt, ref, score):

- **sent:** WMT22 MQM single segments (en-de, en-ru, zh-en) (Freitag et al., 2022); 59,144 rows.
- **agg:** the same segments, concatenated in windows of $k \in \{2, 3, 4, 6\}$ consecutive same-document segments; the window score is the mean segment score; 160,377 rows.
- **native:** WMT25 general-MT human evaluation (Kocmi et al., 2025); the scored unit is a whole document ($k=0$); only documents with a reference are kept, so the reference-based and reference-free arms train on identical rows; 42,558 rows.

Scores are z-normalised per (source, LP) and sigmoid-squashed; this is monotone, so rank correlations are unaffected. Splits are document-disjoint.

4.2 Mixes and arms

Each mix holds exactly 24,000 rows. The sentence fraction $f \in \{0, 10, 20, 40, 60, 80, 100\}\%$ sets the share of single sentences; the remaining mass is split 50/50 between aggregated windows and native documents. Two ablations pin the origin of the long mass at $f=0$: *frac000nat* (all native) and *frac000agg* (all aggregated). Pools are shuffled once (seed 42) and mixes take prefixes, so mixes differ only by prefix length. Crossed with two base metrics (WMT22-COMET-DA, WMT22-COMETKIWI-DA) and two encoder regimes (unfrozen vs. frozen encoder), this gives $2 \times 9 \times 2 = 36$ arms.

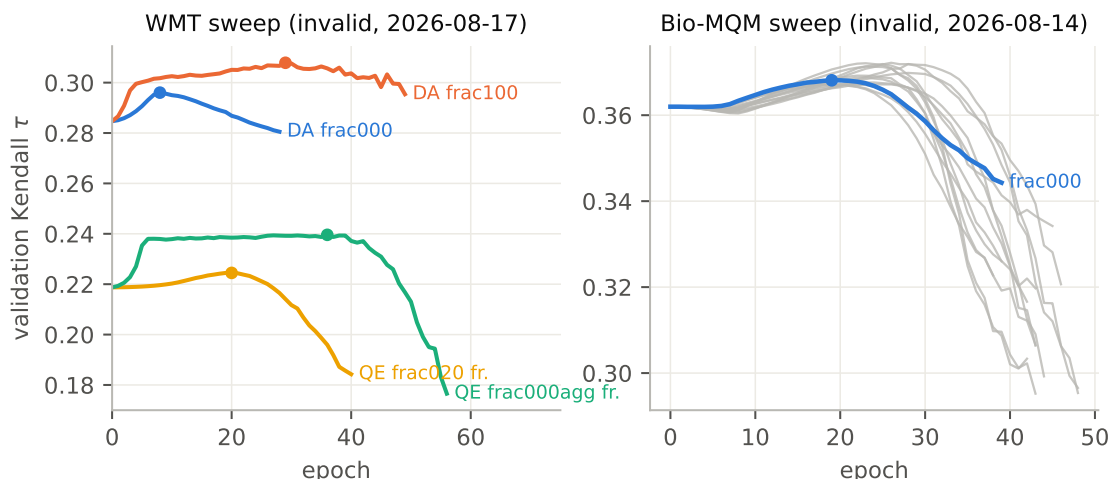


Figure 2: Validation Kendall τ during the two invalid sweeps. Dots mark peaks; every run ends exactly 20 evaluations later (early-stopping patience). Left: representative WMT arms. Right: all Bio-MQM arms (grey), one highlighted.

A known confound. The sent/agg pools cover 3 language pairs, the native pool 13, with almost no overlap. Moving f therefore also moves the language distribution. We keep this explicit and analyse per language pair where it matters.

4.3 The first sweep did not measure composition

A first 36-arm sweep (2026-08-17) produced validation curves that rise and then fall (Figure 2), and an arm ranking that turned out to be an artefact. The cause was in file handling, not in the mixes. COMET selects one training file per epoch (`train_data[epoch % len]`), and the trainer never rebuilt the dataloader. Passing one CSV per language pair therefore trained every arm on *its first file only*: between 205 and 6,940 rows instead of 24,000. Two fatal confounds followed. Training budgets differed by a factor of ~ 34 across arms. And the arms trained on different language pairs. An earlier Bio-MQM version of the sweep (2026-08-14) had the same defect, hidden there by a constant file count per arm.

The bell shape itself has a second, independent cause: early stopping monitored validation Kendall with patience 20 on tiny epochs, so *every* run ended exactly 20 evaluations past its peak, deep into overfitting. The Bio-MQM runs lose up to 0.076 Kendall from peak to final checkpoint.

Corrected protocol. Every arm now trains on its mix’s single concatenated CSV: one epoch is the whole 24,000-row mix in every arm. The budget is fixed grid-wide: max 6 epochs ($\leq 4,500$ optimizer steps at effective batch 32), validation once per epoch, patience 3, and model selection by best validation Kendall, never by the last checkpoint. Three guards prevent regressions: the config generator refuses multi-file training data; the training library warns when files would be ignored; evaluation caches are invalidated by checkpoint fingerprint. Checkpoints from before the correction are quarantined and excluded.

4.4 Evaluation lenses

- **Per- k correlation.** Kendall τ against human scores on: WMT22 sentence and aggregated-window sets ($k \in \{1, 2, 3, 4, 6\}$); natively long WMT25 document sets; and held-out WMT23/24 paragraph sets never seen in training.

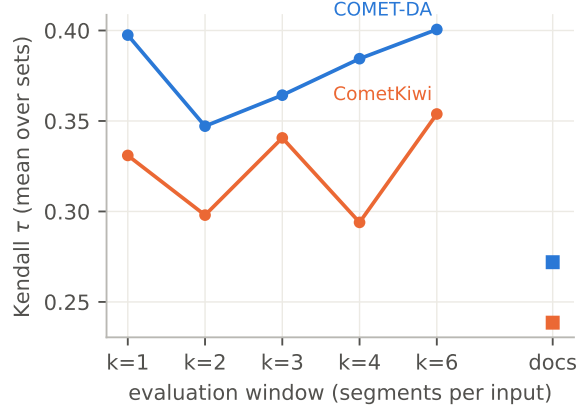


Figure 3: Baseline Kendall τ by evaluation window. Lines: WMT22 aggregated windows. Squares: natively long documents (WMT23–25), the hard case.

- **MetaDocEval** (Dahan et al., 2026). Contrastive accuracy $P[\text{score}(\text{orig}) > \text{score}(\text{perturbed})]$ over nine document-level perturbations, scored with SLIDE-style windows (Raunak et al., 2024) of $w \in \{1, 3, 6, 9\}$ segments. **sentence_splitting** is quality-preserving by design; its “accuracy” is a false-positive rate. Two categories are smaller than the upstream README advertises (verified against the data); all results use measured counts.
- **Block alignment** (Section 2) with the retrained encoders: does mix training restore single-error detection in long blocks?

4.5 Baselines

Figure 3 shows the released metrics under the per- k lens. Both track human judgment best on single sentences ($\tau = .397$ for COMET-DA, $.331$ for CometKiwi, mean over WMT22 pairs). On natively long documents both drop sharply ($.272$ and $.239$). Aggregated windows sit in between and recover with k ; this is expected, since averaging segment scores denoises the target.

On MetaDocEval (Figure 4), both baselines detect structural perturbations near ceiling at $w=1$ and dilute as context grows. For the lexical categories (tense, lexical consistency, conjunction), COMET-DA starts higher and *decays* with w ; no baseline genuinely improves with context. This is the paper’s central negative result, and the question for the sweep is whether any retrained arm climbs instead of decaying.

4.6 Corrected sweep: first wave

Following the correction, we first train the eight arms that answer the headline question: $f \in \{0, 100\} \% \times \{\text{DA}, \text{QE}\} \times \{\text{frozen}, \text{unfrozen}\}$. Startup checks confirm each arm loads exactly one 24,000-row file and the shared budget. Under the corrected budget no validation curve is bell-shaped: every arm improves monotonically through epoch 5, and the best checkpoint is the last one.

Per- k correlation. Table 3 shows the held-out lens. Every arm beats its base metric at every k . The composition effect is small but goes the expected way: the long-only arm is best on native documents, the sentence-only arm is best on single sentences. At this budget there is no catastrophic forgetting: even $f=0$ *gains* $+0.011$ at $k=1$ over the base. Frozen arms track unfrozen arms closely, a little lower; for QE the frozen gap is larger. On the selection-coupled validation lens the document contrast is wider: $\tau(k=0) = .259$ for da-frac000 vs. $.234$ for da-frac100 and $.216$ for the base.

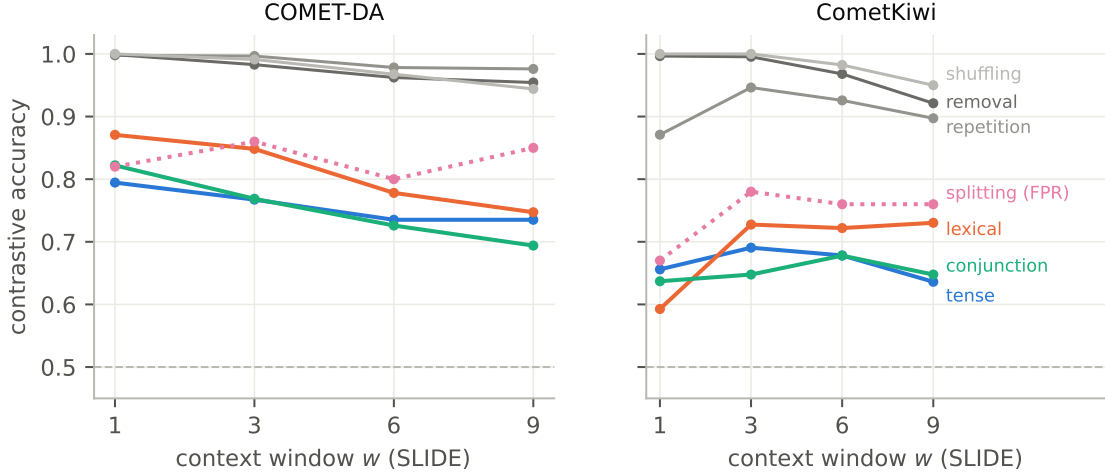


Figure 4: MetaDocEval baselines: contrastive accuracy vs. context window w , micro-averaged per category. Grey: structural perturbations. Dotted: splitting, a false-positive rate. Dashed line: chance.

arm	WMT22 windows					docs
	$k=1$	$k=2$	$k=3$	$k=4$	$k=6$	$k=0$
COMET-DA base	.397	.347	.364	.384	.401	.272
frac000	.408	.360	.378	.408	.430	.288
frac000-frozen	.406	.358	.376	.403	.419	.284
frac100	.410	.361	.378	.404	.421	.284
frac100-frozen	.407	.356	.375	.399	.413	.283
CometKiwi base	.331	.298	.341	.294	.354	.239
frac000	.362	.317	.353	.313	.379	.247
frac000-frozen	.351	.308	.343	.297	.354	.248
frac100	.380	.329	.356	.322	.379	.249
frac100-frozen	.354	.313	.347	.301	.360	.249

Table 3: First corrected wave, held-out Kendall τ (mean over sets). $k \geq 1$: WMT22 en-de/en-ru/zh-en windows. $k=0$: WMT23/24 paragraphs and WMT25 documents (different language mix — compare within a column, not across k).

MetaDocEval. No arm’s lexical-perturbation accuracy rises with the context window w ; the baseline decay pattern is unchanged. The DA arms sit within ± 0.02 of their base everywhere. The QE arms gain modestly at $w=1$ on the lexical categories (tense $.658 \rightarrow .703$ for frac100, conjunction $.637 \rightarrow .678$) but still dilute as w grows. The splitting false-positive rate worsens slightly with retraining (DA: $.82 \rightarrow .84$ at $w=1$; QE: $.67 \rightarrow .74-.78$). So the correlation gains above are score-level, not discourse-level: at this budget, long-text training does not buy detection of document-level errors.

Block alignment. The retrained DA encoders move block-xSIM++ detection only marginally: at $k \in \{3, 4, 5\}$ both arms stay below chance (frac000: $.48/.42/.36$; frac100: $.50/.43/.37$; base: $.47/.40/.34$). The gain is small but directionally consistent — positive in all 16 (language, k) cells, and slightly *larger* for the sentence-only arm. Continued training on human-scored data helps the encoder a little; training on long text specifically does not. Encoder-level length robustness is not restored, consistent with the tiny encoder learning rate (5×10^{-7}) leaving the representation nearly unchanged.

Pipeline audit. Prompted by the counter-intuitive validation curves, we audited the long-text data path end to end. Three findings matter for interpretation. First, a genuine defect: CometKiwi concatenates `mt` and `src` into one sequence truncated at 512 tokens, while our length filter (480 tokens) is applied per side. As a result 7.8% of native and 10.7% of $k=6$ training rows overflow for the QE arms, losing on average $\sim 30\%$ of the source; on the WMT25 evaluation documents the loss reaches $\sim 52\%$ of the source on the 7.1% of rows affected. The DA arms are unaffected (sides are encoded separately). QE-vs-DA contrasts on long inputs must be read with this asymmetry in mind. The filter is now concatenation-aware (rows must also satisfy $\text{src} + \text{mt} \leq 508$ tokens); the wave is being re-run on rebuilt pools. Second, the training-time `val_kendall` is a single Kendall pooled over all 21,988 validation rows, mixing pools and score scales; it is dominated by $k=1$ rows and is not a long-text metric. Third, the aggregated windows, the native pool’s score orientation, and the document-disjoint splits check out (window contiguity has one latent gap affecting 11/72,447 zh-en windows).

4.7 Analysis plan

Pre-specified, per arm and lens: $\Delta\tau$ at $k=1$ vs. the arm’s own base metric (forgetting); the smallest f that recovers baseline sentence-level τ (restoration point); the slope of τ vs. $\log_2 k$ (length profile); frozen-vs-unfrozen contrast (does the effect live in the encoder?); DA-vs-QE contrast (does reference anchoring matter?); MetaDocEval accuracy vs. w (real discourse competence vs. rescaling). Uncertainty: bootstrap over documents, never over windows, 1,000 resamples.

5 Discussion

Three results are already stable. First, COMET’s length problem is not (only) a data problem: the encoder itself stops separating a single error once the input grows, even with meaning-free filler. Second, cheap fixes at the fine-tuning level are weak: in-domain adaptation barely moves representations, and its correlation gains are calibration, not ranking. Third, measuring the effect of training composition is easy to get wrong: our first sweep silently trained every arm on a fraction of its data, and its “findings” reproduced the budget ranking. Constant-budget design, single-file epochs and best-checkpoint selection are prerequisites, not details.

The first corrected wave already answers the headline question in part. Six epochs of continued training improve human correlation at every window size, whatever the mix. The composition effect is real but second-order: long-only training helps most on documents, sentence-only training on sentences, and neither costs the other much. What long-text training does *not* buy, at this budget, is document-level competence: MetaDocEval accuracy never rises with context, and the encoder still cannot find one error in a long block. The remaining 28 arms can map the restoration curve between $f=0$ and $f=100$; the deeper question of discourse competence likely needs more than data composition — longer training, document-level supervision, or architectural change.

6 Conclusion

We isolated length as a failure driver of COMET’s encoder, quantified the limits of domain fine-tuning, and ran the first wave of a corrected, constant-budget sweep over training-mix composition. Continued training helps at every length; composition fine-tunes where it helps most; discourse-level competence does not emerge. Next: the remaining 28 arms to map the restoration curve, and a decision on rebuilding the QE data with a concatenation-aware length filter.

Limitations. One seed per arm; bootstrap CIs quantify sampling noise, not seed noise. The sent/agg vs. native pools differ in language coverage; f moves both composition and language distribution. Aggregated windows use mean segment scores as targets; the mean is itself a modelling choice, which MetaDocEval’s contrastive design does not depend on. Matched-core numbers use four languages and 120 cores per language; the layer probe is a reduced grid (de/fr, two filler types). “Native documents” are capped at 480 XLM-R tokens per side by the data filter; truly long documents are outside this study’s range. The QE arms train and evaluate on partially truncated long inputs (the CometKiwi concatenation defect above); their long-input numbers are lower bounds. The native pool’s document-level ESA signal is weak (within-document contrast $\tau \approx .09$ vs. $.26$ for the MQM pools), so native-heavy mixes learn from a noisier target.

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